

Nonlinear Blaschke Circle Dynamics: An Explicit Annulus and Complete Transfer Spectrum

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Abstract

For the real family $B_a(z) = z(z-a)/(1-az)$, $0 \leq a < 1$, we give a complete circle periodic-point census and its nonconstant stability multipliers. An explicit increasing lift proves that every positive iterate has exactly $2^n - 1$ fixed points, while off-circle contraction isolates the two attracting channels. We construct a parameter-dependent annulus on which the angular Perron operator is trace class. Invariant Laurent sections determine its entire nonzero spectrum, including the reflected-channel multiplicity.

Keywords: Blaschke products; periodic orbits; stability multipliers; transfer operators; Fredholm determinants; singular limits.

中文摘要

本文研究实参数有限布拉施克圆周映射，证明所有迭代的不动点数量及本原周期数量，并记录随轨道变化的稳定性乘子。严格递增的提升给出完整性，圆内与圆外的收缩则隔离两个吸引通道。进一步构造显式参数环域，证明角密度转移算子属于迹类，并从不变洛朗有限维子空间确定全部非零谱及其重数。
关键词：布拉施克乘积；周期轨道；稳定性乘子；转移算子；弗雷德霍姆行列式；奇异极限。

1 Object, provenance, and an all-period orbit theorem

The time variable is integer iteration of $\tau_a = B_a|_{\mathbb{T}}$, where $\mathbb{T} = \{z : |z| = 1\}$ and $0 \leq a < 1$. All complex phases below retain their sign. The explicit transfer spectra of analytic circle maps are prior work of Slipantschuk, Bandtlow and Just [1], and the finite-Blaschke spectral theorem is due to Bandtlow, Just and Slipantschuk [2]. We give a direct real-family proof and a quantitative finite-section certificate. The nonlinear multiplier atlas differs from the workspace's linear expanding-circle and quadratic inverse-branch systems. This owner distinction is not a claim of literature priority.

Choose the lift $\psi_a(0) = 0$ of $B_a(e^{i\theta}) = e^{i\psi_a(\theta)}$. Logarithmic differentiation gives

$$\psi'_a(\theta) = 1 + \frac{1-a^2}{1-2a\cos\theta+a^2}, \quad \frac{2}{1+a} \leq \psi'_a(\theta) \leq \frac{2}{1-a}. \quad (1)$$

The extrema occur at $\theta = \pi$ and 0 . Thus every map in the chamber is expanding, orientation preserving and of degree two.

Theorem 1 (Complete orbit census). *For every $n \geq 1$ the circle fixed count and primitive cycle count are*

$$F_n = 2^n - 1, \quad P_n = \frac{1}{n} \sum_{d|n} \mu(n/d)(2^d - 1). \quad (2)$$

Every circle fixed point of every iterate is simple. For a primitive cycle γ of length p , its multiplier is $\Lambda_\gamma = \prod_{j=0}^{p-1} \psi'_a(\theta_j) > 1$; its s th repetition has length sp and multiplier Λ_γ^s .

Proof. The lift of the n th iterate satisfies $\psi_{a,n}(\theta + 2\pi) = \psi_{a,n}(\theta) + 2\pi 2^n$. The function $\psi_{a,n}(\theta) - \theta$ has positive derivative by (1) and increases by exactly $2\pi(2^n - 1)$ on a half-open fundamental interval. It meets that many multiples of 2π , each exactly once. Nonzero derivative of the fixed-point equation proves simplicity. The identity $F_n = \sum_{d|n} dP_d$ gives Moebius inversion; the chain rule proves the repetition law. $\square$

Complex conjugation commutes with B_a . It pairs or fixes circle cycles, preserving their multipliers; it does not reverse time. Since the map is two-to-one, there is no globally defined inverse dynamics on the circle. The

orientation is positive, and the real Perron weights have phase zero. Their multiplier dependence cannot be removed when constructing traces.

For $0 < s < 1$ the Blaschke factor satisfies

$$|B_a(z)| \leq q_a(s) := s \frac{s+a}{1+as} < s \quad (|z| \leq s). \quad (3)$$

The maximum modulus of $(z-a)/(1-az)$ on $|z| = s$ is $(s+a)/(1+as)$, as follows by differentiating its squared modulus with respect to $\cos \theta$. Therefore no nonzero interior point is periodic. The reflection identity $B_a(1/\bar{z}) = 1/\overline{B_a(z)}$ gives the same result outside the circle. The only off-circle fixed points of an iterate are 0 and ∞ , with local multipliers $(-a)^n$.

The unweighted Artin–Mazur zeta already follows from (2):

$$\zeta_{\text{AM}}(u) = \exp \sum_{n \geq 1} \frac{F_n u^n}{n} = \frac{1-u}{1-2u} \quad (|u| < 1/2). \quad (4)$$

It is independent of a , whereas the stability weights are not.

Round-zero certificate: complete orbit census, stability, repetitions, orientation, and off-circle separation.

2 An explicit trace-class annulus and the exact spectrum

Put $r = (1+a)/2$, $A_r = \{r < |z| < r^{-1}\}$ and $t = (r + q_a(r))/2$. Then $0 < t < r < 1$. The Hardy norm on this annulus is

$$\|f\|_r^2 = \sum_{m \in \mathbb{Z}} |f_m|^2 (r^{2m} + r^{-2m}).$$

Define the angular-density Perron operator by the local branch sum

$$(\mathcal{P}_a f)(w) = \sum_{B_a(z)=w} \frac{w}{z B'_a(z)} f(z). \quad (5)$$

The branches themselves are not global functions on the annulus. Their sum is single valued. Multiplication by z conjugates the derivative transfer convention of [2] to (5).

Lemma 1 (Trace class from annular extension). *The operator in (5) is trace class on $H^2(A_r)$.*

Proof. Choose $s > r$, $s < 1$ with $q_a(s) < t$. Formula (3) and its reflection put every preimage of $\overline{A_t}$ strictly inside A_r , in $s < |z| < s^{-1}$. The critical points are $c = a/(1+\sqrt{1-a^2})$ and $1/c$ for $a > 0$; their images lie outside $\overline{A_t}$ since $c < a < r$. At zero parameter the critical points are zero and infinity. Consequently all local weights are bounded near the compact target boundary. Cauchy–Schwarz on the Laurent series bounds evaluation on the compact source subannulus. This proves boundedness of the extension $\tilde{\mathcal{P}}_a : H^2(A_r) \rightarrow H^2(A_t)$.

The restriction $J : H^2(A_t) \rightarrow H^2(A_r)$ is diagonal with singular values

$$s_m(J) = \sqrt{\frac{r^{2m} + r^{-2m}}{t^{2m} + t^{-2m}}} \leq \sqrt{2}(t/r)^{|m|} \quad (m \neq 0), \quad s_0 = 1.$$

They are summable. The factorization $\mathcal{P}_a = J\tilde{\mathcal{P}}_a$ proves trace class by the ideal property [3]. $\square$

Theorem 2 (Spectrum including both Hardy channels). *For $0 < a < 1$, the nonzero spectrum consists of 1 with algebraic multiplicity one and $(-a)^k$, $k \geq 1$, each with algebraic multiplicity two. Zero is in the spectrum. The entire Fredholm determinant and traces are*

$$D_a(u) = \det(I - u\mathcal{P}_a) = (1-u) \prod_{k \geq 1} (1 - (-a)^k u)^2, \quad (6)$$

$$\text{Tr } \mathcal{P}_a^n = \frac{1 + (-a)^n}{1 - (-a)^n} \quad (n \geq 1). \quad (7)$$

At $a = 0$, $D_0(u) = 1 - u$, but $\mathcal{P}_0$ is not rank one.

Proof. Since $B_a(0) = 0$, the mean-value formula for positive powers of B_a and complex conjugation for negative powers prove Lebesgue invariance. Hence $\mathcal{P}_a 1 = 1$. A branchwise change of angular variable gives, for $m, k \geq 1$,

$$[z^k] \mathcal{P}_a z^m = [z^m] B_a(z)^k. \quad (8)$$

The Taylor coefficients are real. Negative output modes would require the constant coefficient of $z^m B_a(z)^k$, which vanishes; the constant output mode also vanishes. Because $B_a(z)^k$ has order at least k at zero, (8) is zero for $k > m$, and its diagonal is $(-a)^m$. Conjugation gives the second block.

Thus the spaces $V_N = \text{span}(z^{-N}, \dots, z^N)$ are invariant with the advertised diagonal. For their orthogonal projections Q_N , trace class implies $\|\mathcal{P}_a - Q_N \mathcal{P}_a Q_N\|_1 \rightarrow 0$: first approximate by a finite sum of rank-one maps and use strong convergence on their vectors, then bound the remaining trace norm by boundedness of the projections. Fredholm continuity [3] gives (6) locally uniformly, proving all algebraic multiplicities and excluding additional nonzero eigenvalues. Taking traces of powers gives a geometric series and (7). At zero parameter, (8) yields $\mathcal{P}_0 z^{2m} = z^m$, so the zero-spectral part has not vanished. $\square$

Round-one certificate: explicit annular extension, trace class, both spectral channels, and their entire determinant.

References

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